

Ashutosh Raval

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SUMMARY

Experienced Software Engineer with 6 years of building high-performance, distributed, cloud-native web applications. Designed scalable APIs, real-time streaming pipelines, and analytics-driven platforms handling millions of records. Strong foundation in system design, distributed systems, and building highly available, resilient production services.

SKILLS

Programming Languages: Python, Java (Spring Boot), C# (.NET), JavaScript (ES6+), C++

Web Development: React, Node.js, Django, GraphQL, REST APIs, HTML/CSS, Sass

Database: PostgreSQL, MySQL, SQL Server, MongoDB

Cloud & DevOps: AWS, Pulumi (IaC), Docker, Kubernetes, GitHub Actions (CI/CD), Kafka, Grafana

EXPERIENCES

C3.ai | Software Development Engineer

Feb 2025 – Present

Full-Stack Developer [React, TypeScript, Node.js, Python, Kafka, GitHub Actions, Prometheus, CI/CD]

- Designed and developed scalable full-stack applications using **React** and **TypeScript**, supporting high-throughput Supply Chain domain workflows for **10+ Fortune 500** companies, efficiently handling **20K+** data points daily.
- Designed highly available, fault-tolerant distributed backend services using **Python** and **Node.js**, optimizing latency by **25%** under high concurrency while ensuring reliability at scale
- Led end-to-end **design** and **delivery** of **5** cross-functional initiatives, evaluating technical tradeoffs, driving technical design, and aligning engineering approaches with product requirements to consistently meet sprint commitments in **fast-paced Agile environments**.
- Implemented **real-time distributed** event streaming pipelines using **Kafka**, publishing **domain events** from **450+ sources** across 20 plants to decouple services and drive near-real-time processing across **3 downstream** business applications
- Improved production operational excellence by enhancing observability through structured logging, metrics, and **Prometheus/Grafana** dashboards, reducing incident resolution time by **25%** across critical services
- Drove testing best practices by implementing **unit** and **integration** test suites in Python and **JavaScript**, achieving **20% higher code coverage** and **25% fewer production** defects through automated GitHub Actions **CI/CD** pipelines.
- Led incident response and **root-cause analysis** for production issues, implementing fixes to improve system resilience and uptime
- Integrated LLM APIs to build **3** AI-powered features, using prompt design to improve user interactions and generate insights.

Atos-Syntel | Software Engineer - I

Jul 2017 – Dec 2022

Full-Stack Developer [Spring boot, React, Python, MySQL]

- Developed full-stack web applications using **React** and **Typescript** by following modern frontend patterns, integrating **RESTful** APIs and backend services to support data-intensive operational workflows across **10** customers
- Built **multi-threaded** Python utilities and validation pipelines to validate and reconcile **700GB** of on-premises data during Azure cloud **migration**, ensuring data integrity across distributed systems and **improving QA** efficiency by 20%
- Architected a full-stack application using **Spring Boot** and **React**, designing scalable **RESTful** services and **MySQL** data models deployed on Azure to support **30K+** users and manage **10M+** records reliably.
- Led structured peer **code reviews** emphasizing **clean architecture** and **performance optimization**, reducing post-release defects by **20%** and **improving deployment reliability** across CI/CD pipelines
- Engaged with **20** enterprise **stakeholders** across energy and healthcare domains, aligning **business workflows** with engineering solutions while **mentoring junior developers** and accelerating their transition to independent contributors

EDUCATION

Northeastern University, Boston, MA. Master of Science in Software Engineering Systems

Jan 2023 – Dec 2024

PERSONAL PROJECTS

Dollar Dashboard | [GitHub](#) [React.js, Node.js, Express.js, MongoDB, SCSS, OAuth]

- Built a full-stack financial management platform using the **MERN** stack, following **MVC** architecture for modular, scalable design and clean separation of concerns
- Built secure REST APIs for expense tracking and financial goals using OAuth and JWT authentication with real-time data
- Leveraged **GraphQL** for efficient querying, improving backend performance across financial workflows

Cloud Infrastructure and Automation | [GitHub](#) [AWS, Iac, AMI, CI/CD, CloudWatch, Jest, Load Balancer]

- Deployed 'Dollar Dashboard' application, leveraging **Pulumi (IaC)** to provision **AWS VPCs**, **RDS**, and other resources while adhering to best practices for security and scalability
- Developed a **CI/CD** pipeline using **GitHub Actions** and **Jest**, ensuring continuous integration through automated testing and deploying application updates with **high availability and minimal downtime**
- Designed **Auto Scaling** and **ELB** architecture to handle workloads, integrating **CloudWatch** alarms with **SNS/Lambda** notifications